Hospital Engineering and Management

Lecture 16: Hospital Communication and Information Systems

Dr. Sahaj Saxena

Department of Electrical and Instrumentation Engineering
TIET

Course Code: UBM702

Learning Objectives

By the end of this lecture, students should be able to:

- explain the role of communication and information systems in safe hospital operation;
- identify major hospital communication subsystems and their engineering requirements;
- describe the architecture and modules of a Hospital Information System (HIS/HMIS);
- distinguish HIS, EHR/EMR, LIS, RIS, PACS, pharmacy and billing systems;
- explain interoperability using HL7, FHIR and DICOM at an introductory level;
- recognize networking, availability, cybersecurity, privacy and backup requirements;
- relate communication failures and information-system downtime to patient safety.

Connection with Lecture 15

Lecture 15 Established

Medical gases are safety-critical utilities whose reliability depends on source plants, pipelines, alarms, terminal units and preventive maintenance.

Lecture 16 Extends the Same Systems View

A hospital also depends on a continuous **information and communication infrastructure** connecting people, medical devices, departments and decisions.

$$\text{Patient Care} \rightarrow \text{Communication} \rightarrow \text{Information} \rightarrow \text{Decision} \rightarrow \text{Action}$$

Opening Question: What Happens If Communication Fails?

Scenario

An ICU patient deteriorates. The monitor generates an alarm, a nurse calls the physician, laboratory values are pending, and a CT scan is requested.

Systems involved

- bedside monitoring;
- nurse-call system;
- telephone/VoIP;
- laboratory information system;
- radiology and PACS;
- electronic health record.

If one link fails

- alarm may not reach staff;
- result may be delayed;
- wrong patient may be selected;
- image may not be available;
- treatment may be delayed.

Key Message

Communication infrastructure is part of the clinical safety system.

Hospital Communication and Information Systems

Hospital communication systems transfer **voice, data, images, alarms and commands** among people, departments and devices.

Communication Systems

- telephone and VoIP;
- nurse-call and code-call;
- public-address and paging;
- intercom and emergency alerts;
- wired and wireless networks;
- telemetry and device connectivity.

Information Systems

- HIS/HMIS;
- EMR/EHR;
- LIS and RIS;
- PACS;
- pharmacy and billing;
- asset/maintenance systems.

Why These Systems Matter

- Hospitals generate large volumes of time-sensitive clinical and administrative information.
- Care is multidisciplinary; information must move across departments without losing identity or meaning.
- Many decisions depend on near-real-time results, images, alarms and medication information.
- Modern devices increasingly communicate over hospital IT networks.
- Failure, delay or corruption of information may directly affect patient safety.

Engineering Perspective

A hospital network should be treated like any other critical utility: it needs capacity, redundancy, monitoring, maintenance, security and a tested fallback plan.

Communication Modes in a Hospital

Mode	Examples	Engineering concern
Voice	telephone, VoIP, intercom	availability, intelligibility
Alarm/alert	nurse call, code blue, fire alarm	priority, latency, escalation
Data	HIS, LIS, device data	bandwidth, integrity, security
Image/video	PACS, telemedicine, CCTV	storage, bandwidth, latency
Messaging	SMS, secure apps, paging	acknowledgement, privacy
Control	BMS, access control, device commands	authentication, fail-safe design

Basic Hospital Communication Architecture

Clinical Areas $\leftrightarrow$ Access Network $\leftrightarrow$ Core Network $\leftrightarrow$ Data Centre/Cloud

Clinical Edge

Workstations, bedside devices, phones, access points, nurse-call terminals.

Network Core

Switches, routers, firewalls, VLANs, gateways, network monitoring.

Applications

HIS, LIS, RIS, PACS, EHR, databases, backup and analytics.

Design Principle

Avoid a single point of failure between critical clinical areas and essential applications.

Telephone and VoIP Systems

Traditional functions

- internal extension calls;
- external communication;
- emergency hotlines;
- operator assistance;
- interdepartmental coordination.

VoIP adds

- calls over IP network;
- central call management;
- integration with directories;
- softphones and mobility;
- easier scaling.

Engineering Concern

When telephony depends on the IP network, network power and switching become part of the emergency communication chain.

Nurse-Call System: Purpose

A nurse-call system allows patients and staff to request assistance and communicate urgency.

- patient call from bed or bathroom;
- staff-assist call;
- emergency/code call;
- call indication outside patient room;
- annunciation at nursing station;
- escalation to mobile device or central console;
- call logging for response-time analysis.

Safety Function

The system is not merely a convenience feature; it supports timely response to falls, deterioration and emergencies.

Nurse-Call System: Functional Chain

Patient/Staff Call $\rightarrow$ Room Controller $\rightarrow$ Nurse Station $\rightarrow$ Assigned Staff $\rightarrow$ Acknowledgement

Design Questions

- Is the call priority visible and audible?
- Is the correct bed/room clearly identified?
- What happens if the network link fails?
- Is there local indication during central-server downtime?
- Are bathroom pull cords accessible from floor level?
- Is response time recorded and auditable?

Code Blue and Emergency Communication

- Code systems should provide a simple, rapid and unmistakable method of summoning the emergency team.
- The location of the emergency must be transmitted correctly.
- Escalation should continue until the message is acknowledged.
- Audible announcements may be combined with pagers, phones or secure mobile applications.
- Communication must remain available during local power or network failure.

Human-Factors Principle

Emergency communication should minimize ambiguity, unnecessary steps and dependence on memory.

Public Address, Paging and Intercom

Public Address	Paging	Intercom
General announcements, evacuation messages and public instructions.	Targeted message to a person or response team.	Two-way communication between defined locations.

Engineering considerations:

- intelligibility and background noise;
- zoning of announcements;
- priority override for emergencies;
- backup power;
- periodic testing and speaker coverage.

Wireless Communication in Hospitals

Common wireless technologies include:

- Wi-Fi for workstations, tablets, phones and selected medical devices;
- cellular networks for external communication;
- Bluetooth for short-range device connections;
- RFID for identification and asset tracking;
- RTLS technologies for location and workflow applications.

Important

Wireless design must consider coverage, capacity, roaming, interference, security and the electromagnetic environment of medical equipment.

Wi-Fi Planning in Clinical Areas

- Perform a site survey rather than assuming uniform coverage.
- Design for both **signal strength** and **client density**.
- Consider roaming for mobile clinical devices.
- Avoid channel congestion and co-channel interference.
- Separate guest access from clinical networks.
- Provide redundant controllers/core infrastructure where clinically justified.
- Reassess coverage after renovations or equipment changes.

Example

An ICU may have fewer patients than an OPD waiting area but far more connected devices per bed.

Hospital Network: Wired Infrastructure

- structured cabling from telecommunications rooms to clinical outlets;
- fibre backbone between floors/buildings/data centres;
- managed Ethernet switches;
- redundant uplinks for critical areas;
- labelled patch panels and documented cable routes;
- environmental control and secure access to network rooms;
- UPS-supported active network equipment.

Why Wired Networks Remain Important

They provide predictable bandwidth, low latency and stable connectivity for fixed high-demand or safety-critical systems.

Network Segmentation

Concept

A large hospital network should not operate as one unrestricted broadcast/security domain.

Possible segments

- clinical workstations;
- medical devices;
- administrative systems;
- building-management systems;
- CCTV/access control;
- guest Wi-Fi.

Benefits

- limits unnecessary traffic;
- improves fault isolation;
- supports security policy;
- reduces lateral movement after compromise;
- improves monitoring.

Hospital Information System (HIS/HMIS)

A Hospital Information System integrates clinical, administrative and financial workflows.

- patient registration and unique identity;
- appointment and ADT functions;
- clinical documentation;
- order entry and result reporting;
- billing and insurance;
- pharmacy and inventory;
- operation theatre and bed management;
- reports, dashboards and audit trails.

Core Objective

Right information, for the right patient, at the right place and time, to the authorized user.

Common HIS Modules

Module	Typical function
Registration/ADT	patient identity, admission, discharge, transfer
OPD/IPD	clinical encounter documentation
Order entry	requests for laboratory, imaging, medications
Billing	charges, payments, claims
Pharmacy	prescription, dispensing, stock
Inventory	stores, procurement, consumption
Bed management	occupancy and transfers
MRD	records, coding, retention and release
Analytics	quality, finance and operational dashboards

EMR versus EHR

EMR

Electronic Medical Record usually refers to the digital clinical record maintained within a provider or organization.

EHR

Electronic Health Record emphasizes a longitudinal digital health record that can support information exchange across care settings.

Do Not Confuse Digitization with Integration

A scanned paper record is digital, but it is not necessarily structured, searchable or interoperable.

Patient Identity: A Fundamental Safety Issue

- A unique hospital identifier links orders, results, images, medications and billing.
- Duplicate records can fragment clinical history.
- Wrong-patient selection can lead to wrong orders or wrong documentation.
- Matching should use multiple demographic identifiers and controlled workflows.
- Bed number alone is never a reliable patient identity.

Engineering Lesson

Data integrity begins with correct identity. A technically perfect network cannot compensate for a wrong-patient record.

Admission–Discharge–Transfer (ADT)

ADT events communicate where the patient is and what encounter is active.

- registration/admission creates or activates an encounter;
- transfer updates location and responsible unit;
- discharge closes the encounter;
- downstream systems may use ADT information to associate orders and results correctly.

Example

A laboratory or radiology system may receive patient demographics from the HIS rather than asking staff to re-enter them manually.

Laboratory Information System (LIS)

- receives laboratory orders;
- supports specimen accessioning and barcode labels;
- interfaces with analyzers;
- records quality-control and result status;
- supports validation and authorization;
- sends verified results to the HIS/EHR;
- tracks turnaround time and pending work.

Workflow

Order $\rightarrow$ Specimen $\rightarrow$ Analyzer $\rightarrow$ Validation $\rightarrow$ Result $\rightarrow$ Clinician

Radiology Information System (RIS)

RIS commonly supports:

- imaging order receipt;
- patient scheduling;
- modality worklists;
- examination status tracking;
- radiologist reporting;
- report distribution;
- billing/operational information.

Integration

RIS manages the radiology workflow while PACS primarily manages imaging objects and their retrieval/display.

PACS: Picture Archiving and Communication System

- stores and retrieves medical images;
- distributes images to diagnostic workstations and clinical viewers;
- maintains image metadata and study association;
- supports short-term and long-term image access;
- integrates with modalities, RIS and EHR viewers.

Typical Chain

Imaging Modality $\rightarrow$ PACS/VNA $\rightarrow$ Diagnostic Workstation $\rightarrow$ Clinical Viewer

DICOM: Imaging Interoperability

DICOM

Digital Imaging and Communications in Medicine is the principal standard used for communication and management of medical imaging information.

DICOM defines mechanisms for:

- image objects and metadata;
- patient/study/series organization;
- image transfer and storage;
- modality worklists;
- query/retrieve;
- structured reports and related imaging services.

Current Standard

DICOM is continuously maintained; implementations should refer to the current published edition and applicable supplements/change proposals.

Pharmacy Information System

- receives prescriptions/orders;
- supports formulary and drug master data;
- checks stock and dispensing status;
- may support dose, allergy and interaction alerts;
- records batch/expiry information where configured;
- interfaces with billing and inventory systems;
- can integrate barcode medication workflows.

Safety Point

Medication software should support clinical work without encouraging alert fatigue or unsafe overrides.

Bedside Device Connectivity

Examples of network-connected clinical devices include:

- patient monitors;
- ventilators;
- infusion pumps;
- anaesthesia workstations;
- dialysis machines;
- point-of-care analyzers.

Possible functions include central monitoring, device status, alarm forwarding and transfer of measurements to clinical records.

Important

Connectivity does not remove the need for clinical verification, alarm management and local device safety functions.

Interoperability: Why It Is Needed

Without interoperability, staff may repeatedly type the same patient information into multiple systems.

Problems with isolated systems

- duplicate data entry;
- transcription errors;
- inconsistent identifiers;
- delayed results;
- fragmented records.

Interoperability enables

- data exchange;
- common identifiers;
- workflow automation;
- decision support;
- longitudinal records.

HL7 Messaging: Introductory View

HL7 standards support exchange of healthcare information among systems.

- Traditional HL7 Version 2 messaging is widely used for event-driven exchange.
- Messages may carry ADT, orders, observations and other clinical information.
- Interfaces require agreed field mapping, codes, identifiers and acknowledgement handling.

Example Concept

Patient admitted in HIS $\rightarrow$ ADT message $\rightarrow$ LIS/RIS receives updated demographic and encounter information.

FHIR: Modern Health Data Exchange

FHIR

Fast Healthcare Interoperability Resources (FHIR) is an HL7 standard that represents health information as modular resources and supports modern web-based exchange patterns.

Examples of resources include:

- Patient, Practitioner, Organization and Location;
- Observation and DiagnosticReport;
- ServiceRequest and MedicationRequest;
- ImagingStudy and Device;
- Encounter and Appointment.

Status in 2026

FHIR R5 remains the current published release; the R6 sequence is still in development/ballot publication.

Interface Engine

Purpose

An interface engine acts as a controlled integration layer among multiple hospital systems.

It may perform:

- message routing;
- field transformation;
- protocol conversion;
- filtering and enrichment;
- queueing and retry;
- acknowledgement tracking;
- interface monitoring and error logs.

Operational Requirement

Interface failures must generate actionable alerts; otherwise the hospital may believe data are flowing when they are not.

A Typical Integrated Workflow

Registration $\rightarrow$ Physician Order $\rightarrow$ LIS/RIS $\rightarrow$ Device/Modality $\rightarrow$ Result $\rightarrow$ EHR

Integration Benefit

Patient demographics and orders can move electronically, reducing repeated manual entry and improving traceability.

But Integration Adds Dependency

A failure in the interface, network, master data or identity workflow can propagate across several departments.

Medical Records Department and Digital Records

Digital records change the work of the Medical Records Department but do not eliminate its governance role.

- record completeness;
- indexing and coding;
- retention and archival;
- release of information;
- access authorization;
- legal/medico-legal preservation;
- data-quality monitoring;
- audit support.

Data Quality Dimensions

Useful hospital data should be:

- **accurate** – reflects the real clinical/administrative event;
- **complete** – required information is present;
- **timely** – available when needed;
- **consistent** – same meaning across systems;
- **traceable** – origin and modifications can be audited;
- **valid** – conforms to permitted values and formats.

Key Idea

Poor data quality is a patient-safety issue, not only an IT issue.

Audit Trails

An audit trail records significant system activity such as:

- user login and logout;
- viewing a sensitive record;
- creation/modification of documentation;
- order entry and cancellation;
- result verification;
- medication changes;
- administrative configuration changes.

Purpose

Audit trails support accountability, incident investigation, security monitoring and legal/quality review.

Availability: How Much Downtime?

For an information service:

$$A = \frac{\text{Uptime}}{\text{Total time}}$$

Availability	Approx. downtime/year	Interpretation
99%	3.65 days	inadequate for many critical services
99.9%	8.76 hours	moderate availability
99.99%	52.6 minutes	high availability

Important

High availability requires architecture, maintenance, monitoring and operational discipline—not simply a powerful server.

Redundancy for Hospital IT

Possible redundancy layers include:

- dual power supplies and UPS;
- redundant core switches;
- multiple network paths;
- clustered/virtualized servers;
- replicated databases;
- redundant storage;
- multiple Internet/WAN links;
- secondary data centre or disaster-recovery site.

Design Principle

Redundancy is useful only when failover is tested and the common failure modes are understood.

Backup versus High Availability

High Availability

Keeps a service running or restores it rapidly after component failure.

Backup

Preserves recoverable copies of data after deletion, corruption, ransomware or disaster.

Key Distinction

A mirrored corrupted database can remain highly available while still being unusable. Backup and recovery are separate requirements.

Recovery Objectives

Two useful planning terms are:

- **Recovery Time Objective (RTO):** target time to restore an application/service;
- **Recovery Point Objective (RPO):** maximum acceptable amount of data loss measured in time.

Illustrative Example

If an LIS has RTO = 1 hour and RPO = 15 minutes, the recovery design should aim to restore service within one hour and limit unrecoverable data to approximately the last 15 minutes.

Note

RTO and RPO are planning targets; they must be validated through actual recovery exercises.

Downtime Procedures

Every critical digital workflow requires a practical downtime plan.

- How will patients be registered?
- How will orders be placed?
- How will results be communicated?
- How will medications be prescribed and administered?
- How will patient identity be verified?
- How will data generated during downtime be reconciled later?

Engineering Lesson

Resilience includes a safe manual fallback, not only redundant servers.

Cybersecurity: Why Hospitals Are Vulnerable

Hospitals combine:

- valuable personal and health information;
- thousands of users with different roles;
- legacy and modern devices;
- continuous 24/7 operational requirements;
- remote vendor support;
- many interconnected applications;
- clinical urgency that can conflict with strict security controls.

Patient-Safety Perspective

Cybersecurity failures can affect confidentiality, but also availability and integrity of clinical services.

Security Objectives: CIA Triad

Confidentiality

Only authorized persons can access protected information.

Integrity

Information remains accurate and unauthorized changes are prevented or detected.

Availability

Systems and data remain accessible when clinically required.

$$\text{Safe Digital Healthcare} = \text{Confidentiality} + \text{Integrity} + \text{Availability}$$

Identity and Access Management

Good access control includes:

- unique user accounts;
- role-based permissions;
- strong authentication;
- multifactor authentication where appropriate;
- rapid removal of access after staff separation;
- periodic review of privileges;
- privileged-account control;
- audit logs for sensitive actions.

Never

Shared generic user accounts should not be normal practice for clinical documentation because they destroy accountability.

Medical Device Cybersecurity

Network-connected medical devices require joint attention from biomedical engineering and IT.

- maintain an accurate device inventory;
- know software/firmware versions;
- document network addresses and dependencies;
- control remote vendor access;
- apply vendor-supported security updates;
- segment device networks where appropriate;
- monitor unusual traffic;
- plan compensating controls for unsupported legacy devices.

Patch Management in a Hospital

Challenge

In ordinary IT, a vulnerable computer may be patched immediately. In healthcare, the affected system may be a validated clinical workstation or connected medical device that cannot be interrupted without planning.

Therefore:

- assess vulnerability severity and clinical risk;
- follow manufacturer/vendor guidance;
- test where necessary;
- schedule safely;
- maintain rollback/recovery capability;
- document exceptions and compensating controls.

Ransomware Scenario

Scenario

Users suddenly cannot access EHR, PACS or shared files. A ransom message appears on multiple systems.

1. protect patient care and activate downtime procedures;
2. isolate affected systems according to incident-response plan;
3. preserve evidence and notify the response team;
4. assess whether medical devices and critical infrastructure are affected;
5. restore from trusted, tested backups only after containment;
6. reconcile downtime data carefully.

Lesson

Cyber incident response is a clinical continuity process as well as an IT process.

Privacy and Minimum Necessary Access

- Clinical information should be accessible to authorized users based on legitimate care/operational need.
- Screens should not expose unnecessary patient information in public areas.
- Portable devices require physical and logical protection.
- Data exports, removable media and messaging need controls.
- Sensitive records may require enhanced auditing.
- Privacy training must include day-to-day behaviour, not only policy documents.

Data Centre Requirements

A hospital data centre/server room typically requires:

- reliable normal and UPS-supported power;
- generator-backed essential supply;
- controlled temperature and humidity;
- appropriate fire detection/suppression strategy;
- access control and CCTV;
- redundant network links;
- cable management and labelling;
- environmental monitoring;
- water-leak protection and housekeeping.

UPS and Power for IT Systems

- Network switches, servers, storage and communication controllers should have suitable ride-through power.
- UPS autonomy bridges short outages and generator start/transfer time.
- Critical telecom rooms also need protected power, not only the central data centre.
- Battery condition and load should be monitored.
- Power-path maintenance should be planned without creating hidden single points of failure.

Connection to Lecture 12

Information-system availability depends directly on the hospital electrical architecture.

Capacity Planning: Network Bandwidth

A simple engineering estimate is:

$$B_{peak} = \sum_{i=1}^{n} N_i \, b_i \, c_i$$

where N_i is number of users/devices of type i, b_i is nominal bandwidth demand, and c_i is a concurrency factor.

Caution

Actual design must also consider burst traffic, imaging transfers, backups, protocol overhead, redundancy and future growth.

Numerical Example: Bandwidth

Suppose a clinical floor has:

- 50 workstations, estimated 0.8 Mbps each at 40% concurrency;
- 30 Wi-Fi clinical devices, 0.5 Mbps each at 50% concurrency;
- 5 imaging viewers, 8 Mbps each at 60% concurrency.

Estimated simultaneous application traffic:

$$B = (50)(0.8)(0.4) + (30)(0.5)(0.5) + (5)(8)(0.6)$$

$$B = 16 + 7.5 + 24 = 47.5 \text{ Mbps}$$

Interpretation

This is only a preliminary traffic estimate; engineered capacity should include headroom and non-user traffic.

Storage Planning for Medical Imaging

A preliminary annual storage estimate can be expressed as:

$$S_{annual} = N_{studies} \times S_{avg} \times F$$

where F includes replication, metadata, overhead and growth reserve.

Example

50,000 studies/year at average 250 MB/study require about 12.5 TB of raw annual image data before replication, backups and growth allowance.

Important

CT, MR, digital pathology and advanced imaging can change storage demand dramatically.

Telemedicine Infrastructure

Telemedicine may combine:

- video consultation;
- image/document sharing;
- remote monitoring;
- specialist review;
- digital consent and scheduling;
- integration with EHR/HIS.

Engineering requirements include bandwidth, camera/audio quality, privacy, endpoint security, identity verification and reliable documentation.

CCTV and Security Systems

CCTV is part of hospital physical-security infrastructure.

- entrances and exits;
- public corridors and waiting areas;
- parking and perimeter;
- selected restricted zones;
- pharmacy/stores where justified;
- equipment rooms and critical infrastructure access points.

Privacy Principle

Camera placement should respect clinical privacy and local policy/legal requirements; surveillance is not a substitute for appropriate staffing or access control.

Access Control Systems

Electronic access control may use:

- cards/badges;
- PINs;
- biometric methods;
- mobile credentials;
- controlled door interlocks.

Common protected areas: data centre, pharmacy, blood bank, CSSD sterile store, medical-gas plant, electrical rooms and selected clinical areas.

Engineering Requirement

Door behavior during fire alarm, power loss and emergency override must be explicitly designed.

RTLS and Asset Tracking

Real-Time Location Systems can support:

- locating mobile medical equipment;
- patient/staff flow studies;
- workflow optimization;
- preventive-maintenance search efficiency;
- temperature/location monitoring for selected assets.

Biomedical Engineering Use

Reducing time spent searching for infusion pumps or monitors can improve maintenance productivity and equipment utilization.

Alarm Integration and Alarm Fatigue

Integrated systems may forward alarms from devices to central stations or mobile endpoints.

- only clinically actionable alarms should be escalated;
- alarm priority should be preserved;
- message receipt/acknowledgement should be clear;
- delay introduced by middleware must be understood;
- system failure should not disable the device's local alarm;
- alarm policies require clinical governance.

Key Principle

More alarms do not automatically mean more safety.

Clinical Communication Apps

Secure mobile communication can support:

- team messaging;
- escalation of urgent tasks;
- on-call communication;
- image/document sharing;
- alerts from hospital systems.

Do Not Use Consumer Convenience as the Design Criterion

Clinical communication requires governance for identity, access, privacy, retention, auditability and downtime.

Building Management System and IT Interface

BMS supervises engineering utilities such as:

- HVAC status;
- temperature/humidity;
- electrical metering;
- water systems;
- alarms and plant status.

Important Boundary

BMS is an operational-technology system. Integration with enterprise IT should be controlled because compromise or failure can affect physical hospital infrastructure.

Hospital Dashboards and Analytics

Data from multiple systems can support:

- bed occupancy;
- OPD waiting time;
- laboratory turnaround time;
- equipment downtime;
- medication-stock alerts;
- infection-control indicators;
- energy and utility consumption.

Engineering Value

A dashboard is useful only when the underlying data definitions are reliable and the metric leads to action.

Key Performance Indicators for Digital Systems

Possible KPIs include:

- application uptime;
- network availability;
- interface-message failure rate;
- mean time to acknowledge critical IT alarms;
- backup success rate;
- restore-test success rate;
- average help-desk response/resolution time;
- percentage of critical assets with current inventory/security status.

Change Management

Changes to hospital information infrastructure should be controlled.

1. define the requested change;
2. assess clinical and technical risk;
3. plan testing and rollback;
4. obtain appropriate approval;
5. schedule implementation;
6. communicate to affected users;
7. verify post-change performance;
8. document the final configuration.

Reason

An apparently minor software or network change can affect multiple clinical workflows.

Configuration Management

Maintain controlled records of:

- network topology;
- IP address/VLAN allocation;
- server and application versions;
- interface mappings;
- medical-device network dependencies;
- user/role models;
- firewall rules;
- backup/recovery configuration;
- vendor contacts and support agreements.

Principle

You cannot reliably recover a system whose configuration is unknown.

Incident Management

When a communication or information service fails:

1. recognize and log the incident;
2. assess clinical impact and severity;
3. activate downtime/escalation if required;
4. restore safe service;
5. communicate status to users;
6. investigate root cause;
7. implement corrective/preventive action;
8. update documentation and training.

Case Study 1: LIS Interface Failure

Scenario

Laboratory analyzers are functioning, but verified results are not appearing in the EHR.

Possible causes:

- interface engine stopped;
- message queue blocked;
- network path failure;
- patient identifier mismatch;
- incorrect mapping after software update.

Immediate Safety Action

Establish a controlled method for communicating critical results while the interface is restored and then reconcile electronic records.

Case Study 2: PACS Unavailable

Scenario

Emergency CT is completed, but the radiologist cannot access images through PACS.

Discussion:

- Is acquisition complete at the modality?
- Is the network path available?
- Is the PACS archive/viewer down?
- Can the study be viewed locally or through a validated fallback?
- How will the report be documented and reconciled?

Lesson

Imaging continuity requires both modality operation and information-system availability.

Case Study 3: Nurse-Call Failure

Scenario

A ward reports that several patient-call buttons do not generate indication at the nurse station.

1. confirm scope: one bed, one room, one zone or whole system;
2. inspect local device and power/network status;
3. verify annunciator/controller operation;
4. implement increased rounding/manual fallback;
5. repair and function-test every affected call point;
6. document response and root cause.

Case Study 4: Network Outage in ICU

Scenario

The ICU loses network connectivity, but bedside monitors continue displaying locally.

Still available

- local monitor display;
- local alarms;
- manual observation;
- some stand-alone devices.

May be affected

- central monitoring;
- EHR access;
- lab/result retrieval;
- alarm forwarding;
- device-data integration.

Design Lesson

Local safety functions should not depend unnecessarily on enterprise-network availability.

FMEA Example: Hospital Information System

Failure mode	Possible effect	Control
Database unavailable	EHR/HIS inaccessible	clustering, backup, downtime procedure
Interface failure	missing/delayed results	queue monitoring, alerts, fallback
Network core failure	multiple areas disconnected	redundant core and paths
Wrong patient selected	unsafe order/documentation	ID workflow, UI controls, audit
Ransomware	service/data unavailable	segmentation, backup, incident response
Power loss	servers/switches stop	UPS + generator + testing

Role of the Biomedical Engineer

Biomedical engineers increasingly work at the boundary between medical devices and IT.

- specify connectivity requirements during procurement;
- coordinate network onboarding with IT;
- verify device communication after installation;
- maintain device inventory and software versions;
- support cybersecurity risk management;
- investigate device/interface incidents;
- validate data transfer during acceptance testing;
- manage vendor remote access with hospital policy.

Role of the Hospital IT Team

Typical responsibilities include:

- network and server infrastructure;
- identity and access management;
- application administration;
- database, backup and recovery;
- cybersecurity monitoring;
- interface/integration support;
- help desk and incident response;
- user-device management;
- vendor and license management.

Shared Responsibility

Clinical engineering, IT, clinicians, administration and vendors must coordinate for networked medical technology.

Procurement of Networked Medical Equipment

Before purchase, ask:

- What network ports/protocols are required?
- Does the device support DICOM/HL7/FHIR or a proprietary interface?
- What cybersecurity documentation is available?
- How are software updates provided?
- Is remote access required?
- What data are stored locally?
- Can logs be exported?
- What happens during network loss?
- What are the backup and recovery requirements?

Acceptance Testing of Connected Systems

Acceptance should verify more than basic power-on.

- correct patient identity transfer;
- correct order/worklist mapping;
- correct units and parameter names;
- time synchronization;
- network-loss behaviour;
- alarm/notification behaviour;
- user authentication and roles;
- data reaching the intended record;
- audit logging;
- restore/fallback process where relevant.

Time Synchronization

Why It Matters

Hospital systems generate time-stamped orders, results, waveforms, medication records, alarms and audit logs.

If clocks differ significantly:

- event sequence becomes difficult to reconstruct;
- clinical correlation may be misleading;
- logs from different systems cannot be compared reliably;
- incident investigation becomes harder.

Engineering Practice

Use centrally managed time synchronization and monitor significant clock drift.

Master Data and Coding

Interoperability requires consistent meanings, not only connected cables.

- department/location master;
- clinician identifiers;
- test catalogue;
- medication master;
- device identifiers;
- units of measure;
- diagnosis/procedure terminology where applicable.

Example

If one system sends “K” and another interprets it differently, connectivity exists but semantic interoperability fails.

India: Digital Health Context

Indian hospitals increasingly operate within a broader national digital-health ecosystem.

- patient identity and consent-based exchange are becoming more important;
- hospitals may need standards-based interoperability with external health platforms;
- claims, registries and longitudinal health records increase dependence on structured digital data;
- implementation should follow current National Health Authority/ABDM specifications and applicable legal requirements.

Teaching Point

National digital-health architecture does not replace the hospital HIS; it creates additional interoperability and governance requirements.

Standards Landscape

Standard/Framework	Primary relevance
HL7 Version 2	event-driven clinical/administrative messaging
HL7 FHIR	resource-based modern health-data exchange
DICOM	medical imaging information and communication
IEC 80001 family	risk management for IT networks incorporating medical devices
ISO/IEC 27001 family	information-security management
Hospital policies/NABH	governance, records, quality and patient-safety requirements

Principle

Use the edition required by the organization's regulatory, accreditation and contractual context.

Maintenance of Communication Systems

Preventive activities may include:

- nurse-call functional tests;
- emergency phone/hotline tests;
- UPS and battery checks;
- switch/router health monitoring;
- Wi-Fi coverage verification after changes;
- server/storage capacity review;
- backup verification and restore exercises;
- interface queue monitoring;
- firmware/software lifecycle review;
- inspection of telecom/data rooms.

Documentation Required

Maintain updated:

- network diagrams;
- application/interface architecture;
- data-flow diagrams;
- device connectivity inventory;
- downtime procedures;
- backup and recovery plans;
- cybersecurity incident-response plans;
- preventive-maintenance schedules;
- vendor escalation matrix;
- change and incident records.

Classroom Activity 1: Map the Information Flow

Task

For a patient who arrives in the emergency department with chest pain and requires blood tests, ECG, medication and CT imaging, draw the information flow among:

- registration/HIS;
- emergency clinician;
- LIS and laboratory analyzer;
- radiology/RIS/PACS;
- pharmacy;
- billing;
- EHR/clinical record.

Identify three points where an interface failure could delay treatment.

Classroom Activity 2: Design a Nurse-Call Response Chain

For a 30-bed ward, propose:

1. normal patient-call pathway;
2. bathroom emergency-call pathway;
3. staff-assist pathway;
4. escalation after 30–60 seconds without acknowledgement;
5. fallback if the central server/network is unavailable.

Deliverable

One block diagram plus a short explanation of fail-safe behaviour.

Classroom Activity 3: Downtime Tabletop Exercise

Scenario

The hospital HIS, EHR and LIS become unavailable for two hours during peak OPD activity.

Each group should decide:

- how patients will be registered;
- how investigations will be ordered;
- how critical results will be communicated;
- how medications will be documented;
- how paper/downtime data will be reconciled after recovery.

Mini-Project Ideas

1. Network and communication audit of a hospital department.
2. Nurse-call response-time measurement and improvement plan.
3. Downtime-readiness assessment for HIS/LIS/PACS.
4. Biomedical-device connectivity inventory and risk classification.
5. Wi-Fi coverage and device-density study for ICU/ward/OPD.
6. Dashboard for biomedical equipment status and service calls.
7. FMEA of an integrated laboratory or radiology information workflow.

Assignment

Hospital Communication and Information-System Audit

Choose one hospital or hypothetical 200-bed hospital and prepare a 2–3 page engineering note containing:

1. major communication systems;
2. HIS/HMIS modules;
3. one clinical integration workflow;
4. network/availability architecture at a conceptual level;
5. one cybersecurity risk;
6. one downtime scenario and fallback plan;
7. three improvement recommendations.

Summary

- Communication systems carry voice, alerts, data, images and commands across the hospital.
- HIS/HMIS integrates administrative and clinical workflows.
- LIS, RIS, PACS, pharmacy and bedside-device systems must exchange information reliably.
- HL7/FHIR support health-data interoperability; DICOM supports medical imaging information.
- Networks require capacity, redundancy, monitoring and secure segmentation.
- Availability, backup, cybersecurity and downtime procedures are patient-safety requirements.
- Biomedical engineering and IT must jointly manage networked medical technology.

Final Message

A modern hospital is not only electrically connected and medically equipped—it is also an information network whose reliability directly influences care.

Suggested Reading and Standards

- HL7 International, *FHIR Specification* and HL7 Version 2 resources.
- DICOM Standards Committee, *DICOM Standard* (current published edition).
- IEC 80001 family, application of risk management for IT networks incorporating medical devices.
- ISO/IEC 27001 family, information-security management systems.
- National Health Authority, India: current ABDM interoperability and digital-health specifications.
- NABH standards and hospital information-management policies applicable to the institution.

Next Lecture: Admission Department and Medical Records Department

Thank You

Questions and Discussion